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[Original Article]

Expressive rhythm training improves consistency of rhythm production abilities in patients with schizophrenia

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Abstract

Patients with schizophrenia often experience deficits in timing and rhythm-processing abilities, yet the impact of group-based expressive rhythm training on these functions remains unclear. In this study, we examined the effects of short, daily rhythm training sessions on rhythm perception and production in patients with schizophrenia. Participants ($N = 15$) in an acute care ward received an average of 18.6 training sessions (15 minutes each). Rhythm abilities were assessed before and after the intervention using the Harvard Beat Assessment Test (H-BAT). Group-level analyses showed no significant changes in the mean values of H-BAT measures. However, correlation analysis revealed that the association between the two rhythm production subtests strengthened considerably post-training (from $\rho = -0.539$ to $\rho = -0.896$), indicating greater consistency in rhythm production performance. These findings suggest that expressive rhythm training may help stabilize rhythm output in patients with schizophrenia, even when average performance levels do not significantly change. This improvement in internal consistency provides new insights into the potential mechanisms linking schizophrenia pathology to impaired rhythmic processing and may contribute to the development of novel rehabilitation strategies for this population.

Key words : Rhythm training, Schizophrenia, Rhythm perception and production, Harvard Beat Assessment Test, Amusia.

Introduction

Occupational therapy (OT) assists patients by focusing on their activities of daily living. Anecdotal evidence suggests that patients with schizophrenia often struggle with communication, recreational activities (sports, social media, etc.), and tasks requiring precise timing. These difficulties seem largely due to deficiencies in timing or rhythm-processing abilities. Indeed, during OT music activities, these patients frequently find it challenging to synchronize their movements to a musical rhythm[1)]. Timing distortions in patients with schizophrenia have been reported for a long time and have been

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studied through several experimental paradigms[2-8]. These impairments in timing and rhythm processing abilities are thought to affect several other negative symptoms, such as anhedonia and decreased activity. In other words, it is thought that patients with schizophrenia are susceptible to various disorders and influences in their lives due to impairments in their ability to process timing and rhythm. Thus, examining rhythm processing ability in patients with schizophrenia may provide insight into improving negative symptoms and daily activities in patients with schizophrenia.

Previous research has considered the potential difficulty faced by these patients in discerning emotional meanings from various sensory cues, including the intensity and pitch of auditory stimuli[9]. More recently, Honda *et al.* (2023)[10] found a relationship between deficits in musical rhythm perception and production and cognitive impairments in these patients. These findings suggest a potential connection between schizophrenia pathology and impaired musical abilities.

The term 'amusia' describes the loss or decline of musical abilities[11-15]. Expressive amusia, a disorder of musical production, and receptive amusia, a disorder of musical perception, are the two primary forms of this condition[11]. The terms "expressive" and "receptive" are used to represent productive and perceptive musical ability, respectively[16]. Specifically, expressive ability involves the reproduction of musical pitch and rhythm through activities such as singing and clapping, while receptive ability involves the perception of musical pitch and rhythm. Intriguingly, previous research on amusia indicates a dissociation in the ability to perceive and produce musical pitch and rhythm, which is thought to result from the distinct neural pathways for perceiving and producing musical pitch and rhythms[17-20]. Honda *et al.* (2023)[10] showed that there was a dissociation between rhythm perception and production abilities in patients with non-treatment-resistant schizophrenia (non-TRS). The patients with non-TRS showed impairment only in the rhythm perception but not in the production. TRS showed impairments in both rhythm perception and production.

Music therapy, which can be either expressive or receptive, has been employed for patients with brain and psychiatric disorders, and its therapeutic impacts have been explored[21-23]. While several therapeutic applications for patients with schizophrenia have been examined through the lens of music therapy, many aspects remain unclear due to differences in the number of sessions, duration, participants, music utilized, and program content[25,26]. Despite these limitations, recent research has indicated the positive effects of expressive music therapy in group settings[22]. Yet, it remains unclear how expressive rhythm training programs in group settings enhance the rhythm perception and production abilities in patients with schizophrenia.

In this study, Expressive Rhythm Training refers to training that involves the expressive production of rhythm through clapping hands and vocalizing in line with the metronome counts and the timing indicated in the musical score. The Expressive Rhythm Training in this study was adapted from a commercially available rhythm training program called "In-body Metronome" (Rittor Music Co., Ltd., Tokyo, Japan) because of the current lack of standardized music interventions in patients with schizophrenia. A meta-analysis by Jia *et al.* (2020)[26] revealed considerable variability in the session duration, frequency, and total intervention length across studies, with no clear consensus on optimal conditions. Against this backdrop, our protocol was informed by our previous mismatch negativity (MMN) research by Takahashi *et al.* (2023)[1], which employed an auditory oddball task using regularly timed stimuli similar to a metronome. Furthermore, preliminary clinical observations—based on our pilot music-based training—suggested that patients with schizophrenia typically maintain sustained attention for approximately 15 minutes. Taking both empirical evidence and practical feasibility into account, each training session was limited to 15 minutes. This protocol can thus be regarded as a pilot framework to support the future development of standardized rhythm training interventions for this population.

The Harvard Beat Assessment Test (H-BAT) is a tool developed to assess an individual's ability to perceive and produce musical rhythm within a single session[20,27]. In this study, we aimed to investigate how expressive rhythm training in a group setting, using H-BAT, improves the ability of patients with schizophrenia hospitalized in acute care wards to perceive and produce musical rhythms.

Methods

Participants

Consent for this study was obtained from 32 patients with schizophrenia who were admitted to the Fukushima Medical University Hospital between

October 2018 and May 2020 and were prescribed OT. All the participants were of Japanese ethnicity and were diagnosed with schizophrenia based on the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition. The participants of this study did not have any other neurological, neurosurgical, or psychiatric conditions, apart from superficially highlighting head injuries. Out of these 32 patients, 15 were eligible for the study, as 17 could not undergo H-BAT measurements before and after rhythm training.

The participants' demographic data (mean ± standard deviation) included age (39.2 ± 15.8 years), sex (7 males and 8 females), duration of illness (131.8 ± 82.5 months), antipsychotic medication (CP-equivalent/pre : 783.42 ± 597.69 mg, post : 798.29 ± 550.70 mg), Global Assessment of Functioning score (GAF : 27.7 ± 10.3), Clinical Global Impression-Severity (CGI-S pre : 6.2 ± 0.94, post : 5.8 ± 1.14), Clinical Global Impression-Improvement(CGI-I : 3.1±0.91), music performance experience (10.5 ± 5.7 years), Brief Evaluation of Psychosis Symptom Domains[28,29] (BE-PSD "Total" pre : 15.4±6.24, post : 13.3±5.58), and rhythm training participation (18.6 ± 17.9 sessions) (see Table 1).

This study was conducted following the approval from the Ethics Committee of Fukushima Medical University (no. 30141). All participants received thorough explanations of the experiment both verbally and in written form, and we obtained their written informed consent.

H-BAT

Rhythmic ability was assessed using an Apple iPad (Apple, Cupertino, CA, USA) with the H-BAT app for iOS installed. The assessment involved the Music Tapping Test (MTT) and both the perception and production parts of the Beat Interval Test (BIT) within the H-BAT[20,27]. The MTT measures the ability to synchronize with a musical beat. The sound stimuli in the MTT included pop-orchestral, jazz, and rock songs at different tempos (100, 120, and 120 beats per minute ; three genres × three tempos = nine patterns of song). Participants were asked to tap in synchrony with the beat while listening to each song. The tap data in relation to the beat timings were recorded, and the entropy of the relative phase between the tap and beat timings was computed. This computed entropy of the relative phase distribution was considered as the Synchronization Index (SI_{ENT}). The larger the SI_{ENT}, the more locked a participant's tap phase is to the beat timings. The perception part of the BIT (BITperception : BITper) assesses the threshold at which a participant can discriminate changes in beat intervals. The BITper stimuli consisted of a sequence of one pure tone followed by 21 woodblock tones. As the inter-tone intervals of the woodblock tones gradually increased or decreased, participants were instructed to discriminate whether the tempo was getting faster or slower by selecting the "gradually faster" or "gradually slower" button on the iPad. The perception threshold was determined in milliseconds using the psychophysical adaptive two-

Table 1. Participants' demographics

	Mean	*SD*
Age (years)	39.2	15.8
Duration of illness (months)	131.8	82.5
Antipsychotic medication (CP- equivalent : mg) pre	783.42	597.69
Antipsychotic medication (CP- equivalent : mg) post	798.29	550.70
GAF on admission (points)	27.7	10.3
CGI-S pre (points)	6.2	0.94
CGI-S post (points)	5.8	1.14
CGI-I (points)	3.1	0.91
BE-PSD Total pre (points)	15.4	6.24
BE-PSD Total post (points)	13.3	5.58
Music performance experience (years)	10.5	5.7
Rhythm training participation (sessions)	18.6	17.9

Mean and SD calculated from fifteen participants (7 males and 8 females).
GAF : Global Assessment of Functioning score. CGI-S : Clinical Global Impressions Severity. CGI-I : Clinical Global Impressions Improvement.
BE-PSD : Brief Evaluation of Psychosis Symptom Domains.

down one-up staircase method. The lower the threshold, the more accurate the individual is at perceiving changes in beat intervals (see Fujii and Schlaug, 2013 for details)[20]. The production part of the BIT (BITproduction : BITpro) assesses the threshold at which an individual can reproduce changes in the beat intervals. The BITpro stimuli are the same as those in BITper. As the inter-tone intervals of the woodblock tones gradually increased or decreased, participants were asked to tap in time with the tones on the iPad, adapting their tapping to the temporal change as precisely as possible. The inter-tap intervals were analyzed for each trial, and a custom-made algorithm (see Fujii and Schlaug, 2013 for details)[20] determined whether the tempo change of the tap matched that of the stimulus. The production threshold was determined in milliseconds using the psychophysical adaptive two-down one-up staircase method. The lower the threshold, the more accurate the individual is at reproducing changes in beat intervals. For both BITper and BITpro, the thresholds were normalized via a log transformation with a base of two following the methodology of previous research[20,30]. To provide clarity on the rhythm indices used, the following summarizes their measurement targets : SI_{ENT} evaluates how consistently participants synchronize their tapping to a steady musical beat, with higher values indicating better synchronization and temporal consistency. BITper measures the perceptual threshold for detecting changes in beat intervals by requiring participants to judge whether the tempo is getting faster or slower. Lower thresholds indicate greater sensitivity to temporal interval differences. BITpro assesses the production threshold for generating changes in beat intervals, evaluating how precisely participants can tap in synchrony with gradually changing tempos. Lower thresholds indicate better ability to generate fine temporal variations.

A registered OT administered the MTT, BITper, and BITpro to each patient in a quiet room. Alongside the verbal explanation of each test, we conducted two practice trials for each test before carrying out the actual tests. All patients used the same iPad for the H-BAT measurement ; the volume was set to MAX, and earphones were not used.

Rhythm training

We used the "In-body metronome" (Rittor Music Co., Ltd., Tokyo, Japan) for expressive rhythm training, which is designed to improve musical abilities in a group setting. In this training, participants were engaged simultaneously with the same piece of music. They read from a musical score, which detailed counts and clapping timings for each piece of music at various feeling of beats, such as 4, 8, and 16 beats. Participants were instructed to clap their hands and vocalize in line with the counts played over the speakers and the timing as indicated in the musical score. If a participant's timing was incorrect, they were arranged to be seated facing each other to ensure audible and visual feedback was available, encouraging them to correct themselves as required. The songs selected for rhythm training were consistent in each session and were arranged in a manner that gradually increased difficulty from a simple 4-beat to a 16-beat. We chose Japanese nursery rhymes and recently popular songs that all participants were familiar with for the training. The rhythm training was carried out once daily for 15 minutes, five days per week. It took place in an OT room within the ward, with care taken to minimize everyday noise disturbances. Participation in each training session was flexible, depending on each participant's condition.

Statistics

The mean values and standard deviations of SIENT, BITper, BITpro, CGI-S, and BE-PSD were calculated before and after expressive rhythm training. The normality of the difference scores (post−pre) for each measure was assessed using the Shapiro-Wilk test. For variables with confirmed normality ($p > .05$), including BITper, BITpro, SI_{ENT}, BE-PSD "Depression/Anxiety", and BE-PSD "Total Score", paired t-tests were conducted. For variables that did not meet the assumption of normality, including CGI-S and BE-PSD "Psychotic Symptoms", "Disorganized Thinking", "Negative Symptoms", and "Excitement/Mania", the Wilcoxon signed-rank test was used. Effect sizes were calculated for all tests : Cohen's d for normally distributed variables and effect size r for non-normally distributed variables. Corresponding test statistics were also computed and reported for each analysis. Wilcoxon signed-rank tests were used for non-normally distributed variables ; however, post hoc power was estimated using a matched-pairs t-test approximation in G*Power due to the lack of direct support for non-parametric tests.

Correlation coefficients were calculated to evaluate relationships among H-BAT subscales and between H-BAT, CGI-S, and BE-PSD scores. As the Shapiro−Wilk test indicated non-normality for some variables, Spearman's rank correlation coefficient (ρ)

was used for correlation analyses involving those variables. The significance level for all statistical analyses was set at $p < .05$.

Results

H-BAT, CGI-S, BE-PSD

The mean and standard deviation of SI_{ENT} before and after expressive rhythm training changed from 0.32 ± 0.08 to 0.31 ± 0.07, BITper from 1.74 ± 1.25 to 1.48 ± 1.27, and BITpro from 0.25 ± 1.84 to 0.53 ± 0.82. SI_{ENT} ($t = 1.02, p = 0.32, d = 0.27$), BITper ($t = 0.86, p = 0.41, d = 0.22$), and BITpro ($t = -0.70, p = 0.50, d = 0.18$) did not show statistically significant differences before and after the training. Furthermore, post hoc power analyses revealed low power for detecting effects of these sizes, with power values of 0.26 for SI_{ENT}, 0.20 for BITper, and 0.16 for BITpro, indicating limited sensitivity due to the small sample size.

The mean score of the CGI-S decreased from 6.2 ± 0.94 before training to 5.8 ± 1.14 after training. The Wilcoxon signed-rank test revealed a statistically significant difference ($Z = -2.121^*, p = 0.034, r = 0.55$), indicating a large effect size. The subscale scores of the BE-PSD changed as follows (see Table 3 for details). Psychotic Symptoms : 3.2 ± 2.11 to 2.9 ± 1.84 ($Z = -1.89, p = 0.06, r = -0.49$), Disorganized Thinking : 3.3 ± 1.95 to 3.0 ± 1.85 ($Z = -1.89, p = 0.06, r = -0.49$), Negative Symptoms : 3.6 ± 1.45 to 3.2 ± 1.37 ($Z = -2.45^*, p = 0.01, r = -0.63$), Excitement/Mania : 2.3 ± 1.62 to 1.5 ± 1.12 ($Z = -2.46^*, p = 0.01, r = -0.64$), Depression/Anxiety : 3.0 ± 0.75 to 2.7 ± 0.70 ($t = 1.29, p = 0.217, d = 0.33$), Total Score : 15.4 ± 6.24 to 13.3 ± 5.58 ($t = 2.843^*, p = 0.013, d = 0.73$). Post hoc power analyses for the non-significant comparisons showed power values of 0.545 for both Psychotic Symptoms and Disorganized Thinking, and 0.334 for Depression/Anxiety.

Correlation

BITper did not show a significant correlation with BITpro either before or after rhythm training ($\rho = 0.338, p = 0.218$ to $\rho = 0.328, p = 0.232$). In contrast, SI_{ENT} exhibited significant correlations with BITpro both before and after expressive rhythm training : a moderate negative correlation was observed before the training ($\rho = -0.539^*, p = 0.038$), and a strong negative correlation was observed after the training ($\rho = -0.896^{**}, p < 0.001$; see Figure 1 and Table 4). No significant relationship was found between CGI and rhythm processing ability. A significant correlation was found between GAF at admission and BITpro, both before and after training ($\rho = -0.823^{**}, p < .001$ to $\rho = -0.616^*, p = 0.01$). Regarding the relationship between BE-PSD and rhythm processing ability, significant correlations were observed between the negative symptom subscale of BE-PSD and SI_{ENT} both before and after training ($\rho = -0.518^*, p = 0.048$ to $\rho = -0.629^*, p = 0.012$). Although no significant correlation was found between BE-PSD negative symptoms and BITpro before training ($\rho = 0.507, p = 0.054$), a strong positive correlation emerged after training ($\rho = 0.752^*, p = 0.001$).

Discussion

This study aimed to investigate how expressive rhythm training improves the abilities of patients with schizophrenia to perceive and produce musical rhythms. We used the H-BAT, a battery of tests, to assess rhythm perception and production abilities before and after the training. Our group analysis revealed that the mean values of the H-BAT measures did not change significantly after the training (Table 2). Nevertheless, when we assessed the

Table 2. The mean and SD of H-BAT measures pre and post expressive rhythm training.

	Pre	Post
SI_{ENT}	0.32 ± 0.08	0.31 ± 0.07
BITper (Log2 ms)	1.74 ± 1.25	1.48 ± 1.27
BITpro (Log2 ms)	0.25 ± 1.84	0.53 ± 0.82

Mean ± standard deviation (SD). SI_{ENT} represents the synchronization index calculated from the entropy of relative phase distribution between musical beat and tap timings. BITper and BITpro denote the thresholds to perceive and produce beat intervals, respectively, where the values were normalized via a log transformation with a base of two (Log2 ms).

Table 3. The mean and SD of BE-PSD measures pre and post expressive rhythm training.

	Pre	post
Psychotic symptoms	3.2±2.11	2.9±1.84
Disorganized thinking	3.3±1.95	3.0±1.85
Negative symptoms	3.6±1.45	3.2±1.37
Excitement/Mania	2.3±1.62	1.5±1.12
Depression/Anxiety	3.0±0.75	2.7±0.70
Total	15.4±6.24	13.3±5.58

Mean ± standard deviation (SD). BE-PSD score before and after expressive rhythm training.

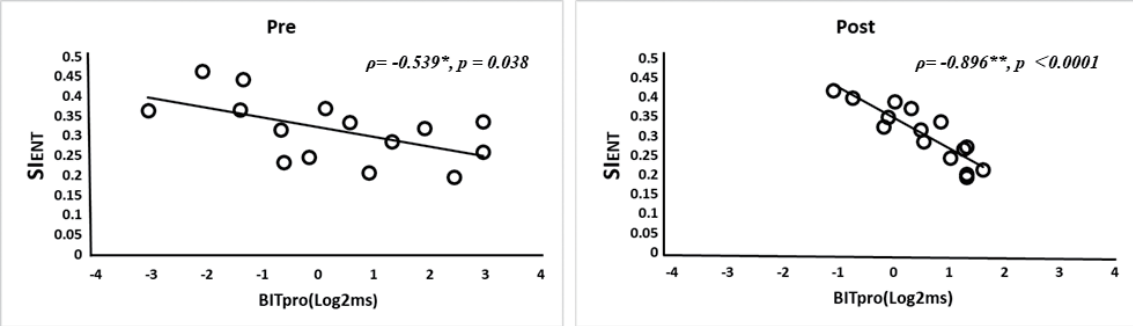

Fig. 1. Correlations between two rhytlun production measures before and after expressive rhytlun training. SIENT represents the synchronization index calculated from the entropy of the relative phase distribution between musical beat and tap timings. BITpro denotes the thresholds to produce beat intervals. Spearman's rank correlation coefficient (ρ) and two-tailed significance probability (p) are shown. $*p < 0.05$, $**p < 0.001$.

Table 4. Correlation coefficients among the H-BAT measures pre and post expressive rhythm training.

	Pre	Post
BITper- BITpro	$\rho = 0.338, p = 0.218$	$\rho = 0.328, p = 0.232$
BITper - SI_{ENT}	$\rho = -0.508, p = 0.053$	$\rho = -0.533^*, p = 0.041$
BITpro - SI_{ENT}	$\rho = -0.539^*, p = 0.038$	$\rho = -0.896^{**}, p < 0.001$

BITper and BITpro denote the thresholds to perceive and produce beat intervals, respectively. SIENT represents the synchronization index calculated from the entropy of the relative phase distribution between musical beat and tap timings. Spearman's rank correlation coefficient (ρ) and two-tailed significance probability (ρ) are shown. $*p < 0.05$, $**p < 0.001$.

correlations among the H-BAT measures, the degree of correlation between the two rhythm production measures (i.e., SI_{ENT} and BITpro) showed a tendency to become stronger after the training ($\rho = -0.539^*, p = 0.038$ to $\rho = -0.896^*, p < .001$) (Table 4). The results suggest that rhythm production abilities become more consistent in patients with schizophrenia after the training (Figure 1). It is noteworthy that, although no statistically significant difference was observed, BITpro scores decreased from 0.25 ± 1.84 to 0.53 ± 0.82 following the training. The rhythm training task employed in the present study emphasized maintaining a steady beat in meters such as quadruple and octuple time. Specifically, participants were required to perceive rhythm embedded within a melody and synchronize their tapping to a consistent pulse. Therefore, it is plausible that the training did not directly impact the ability to produce rhythm (BITpro). This finding suggests a meaningful implication for future research and intervention design : rhythm training programs should incorporate not only externally guided tasks (e.g., following a musical score), but also internally driven tasks that require participants to generate rhythm independently based on their own timing mechanisms.

The CGI-S score significantly decreased after the training, and significant reductions were also observed in the BE-PSD subscales of "Negative Symptoms", "Excitement/Mania", and "Total Score". Because the CGI-S is based on a clinician's overall impression and not limited to symptom-specific assessment, the observed improvements in BE-PSD contributed to the reduction in CGI-S. The question arises as to what factors led to these improvements in psychiatric symptoms. It is important to note that the current study involved a small sample ($N = 15$) of inpatients with schizophrenia, who were concurrently undergoing pharmacological treatment. Nonetheless, previous studies (e.g., Jia *et al.*, 2020[26]) have reported reductions in negative and general psychiatric symptoms following music-based interventions in individuals with schizophrenia. These findings support the potential efficacy of the expressive rhythm training used in this study.

Global functioning at admission was assessed using both the GAF and the CGI. While no significant relationship was found between CGI and rhythm processing ability, GAF scores at admission

were significantly correlated with BITpro both before and after training ($\rho = -.823^{**}, p < .001$; $\rho = -.616^{*}, p = .01$). This suggests that individuals with lower functional status at baseline tended to have poorer rhythm production ability throughout the intervention period.

Furthermore, significant correlations were observed between rhythm processing ability and BE-PSD, particularly between the negative symptom subscale of BE-PSD and SI_{ENT}, both before and after training. Although BITpro was not significantly correlated with negative symptoms before training ($\rho = 0.507, p = .054$), a strong positive correlation emerged post-training ($\rho = .752^{**}, p = .001$). Of particular interest is the relationship between negative symptoms such as avolition and psychomotor retardation and motoric rhythm indicators like SI_{ENT} and BITpro. This aligns with previous research suggesting that musical interventions may positively impact negative symptoms, supporting the view that dynamic, expressive rhythm training can contribute to improvements in motivational and behavioral domains among individuals with schizophrenia.

Mean values of H-BAT measures

The mean values of SI_{ENT}, BITper, and BITpro for patients in this study after the training were 0.31 ± 0.07, 1.48 ± 1.27, and 0.53 ± 0.82, respectively (Table 2). A previous study showed that these mean values from the healthy participants were 0.43 ± 0.04, 0.33 ± 1.41, and −1.28 ± 0.90, respectively[22]. Since the lower SI_{ENT} and the higher BITper and BITpro indicate worse rhythmic performance, all the mean values in this study from the patients appear worse than those of healthy individuals, even after rhythm training. This finding is consistent with the recent study by Honda *et al.* (2023)[10] who reported lower performances in H-BAT measures in patients with schizophrenia. These findings suggest that patients with schizophrenia have difficulty perceiving and producing musical rhythms compared to healthy individuals.

Why didn't we observe any improvements in the mean values of H-BAT measures after the training in the patients? This might be because the duration and frequency of our training weren't sufficient to bring about improvements. For instance, a previous study by Satoh *et al.* (2014)[29] highlighted the positive effects of musical training on elderly individuals by administering music-based exercise training for about 60 minutes per week over a year. Another study by Bhide *et al.* (2013)[30] demonstrated the positive effects of rhythmic training on patients with dyslexia through 19 sessions of 25-minute training spanning 2 months. In contrast, the patients in our study underwent an average of 18.6 rhythmic training sessions, with each session lasting 15 minutes per day. Therefore, the training duration and frequency in our study might have been insufficient. A longer training duration and more frequent sessions may be necessary to observe improvements in the mean values of H-BAT measures.

Improved consistency in rhythm production measures

Notably, the correlation between SI_{ENT} and BITpro showed a marked increase after the training ($\rho = -.539^{*}, p = .038$ to $\rho = -.896^{**}, p < .001$; Figure 1). While direct comparisons are limited due to methodological differences, a previous study reported a correlation of $r = -.512$ among healthy participants[20]. This suggests that the pre-training correlation among patients was comparable to that of healthy individuals. However, the post-training correlation exceeded that of healthy participants. Why did such a strong correlation between the two rhythm production measures emerge after training? We propose at least two explanations for this observed enhancement. First, the training may have improved motor-based rhythmic output. In our intervention, patients were instructed to express rhythm through clapping and vocalization while listening to music. Similarly, both the MTT and BITpro tasks in the H-BAT required rhythmic motor output via tapping to musical stimuli. This overlap in motor demands may have promoted consistency in performance across subtests that share a motor component. Second, rhythm training may have supported sustained attention during motor tasks. Each training session lasted 15 minutes per day, mirroring the approximate duration of the H-BAT assessment. To perform consistently across H-BAT subtests, participants were required to maintain attentional focus throughout the session. Thus, the rhythm training may have strengthened their ability to sustain attention over time, contributing to the heightened consistency observed post-training. Interestingly, BITpro scores increased after training, which may suggest a modest reduction in beat generation precision. However, the strengthened correlation with SI_{ENT} points to a possible improvement in temporal consistency during rhythm production. While BITpro assesses the capacity to spontaneously generate precise rhythmic beats, SI_{ENT} evaluates the ability to maintain a steady tempo. Therefore, rhythm training may have enhanced beat stability rather than mo-

ment-to-moment accuracy.

Furthermore, BITpro performance may be influenced by changes in internal timing strategies or shifts in attentional control. The increased BITpro scores might reflect such adaptive changes rather than a straightforward decline in performance. The improvement in rhythmic consistency, as captured by SI_{ENT}, may also reflect gains in sustained attention and concentration. These cognitive enhancements could extend beyond musical tasks, potentially supporting smoother interpersonal interactions and better behavioral regulation in daily life. Indeed, the observed post-training reduction in CGI-S scores may be linked to such functional improvements. Taken together, these findings suggest that expressive rhythm training may benefit not only motor timing performance but also broader cognitive and functional domains relevant to clinical recovery.

Limitation

There are several limitations in the present study. First, since we focused primarily on the rhythmic abilities of patients with schizophrenia, we did not examine in detail the relationships between rhythm training and clinical symptoms. Investigating how rhythm training affects positive symptoms, negative symptoms, and cognitive dysfunction in patients with schizophrenia would be of great value. Future research should explore these associations in greater depth. Second, this study involved a small sample size ($N = 15$), which may have limited statistical power. The small sample increases the risk of type II errors, potentially overlooking existing effects, and leads to unstable estimates of effect sizes. Therefore, the generalizability of the findings is limited. Larger-scale studies are needed to verify and extend the preliminary results reported here. Third, given the heterogeneity of symptoms and individual cognitive differences among patients with schizophrenia, the optimal duration and effectiveness of rhythm training may vary between individuals, suggesting that a fixed session length may not be ideal for all patients. Moreover, considering the negative symptoms and cognitive impairments characteristic of schizophrenia, maintaining patient motivation and adherence to the intervention can be challenging. Finally, because rhythm training is typically conducted alongside pharmacological and psychosocial treatments, isolating the specific effects of rhythm training remains difficult. These factors highlight the need for individualized treatment approaches and further research to optimize rhythm training for practical clinical application.

Conclusions

We investigated how expressive rhythm training in a group setting might benefit patients with schizophrenia in perceiving and producing musical rhythms. We administered sessions lasting 15 minutes each day to patients in an acute care ward. Using the H-BAT, we found that the degree of correlation between rhythm production measures showed a tendency to become stronger after the training. This result suggests that rhythmic training enhances the consistency of rhythm production abilities in patients with schizophrenia. Our study, conducted in a hospitalized setting, offers valuable insights for the rehabilitation of patients with schizophrenia and will contribute to the future elucidation of the neural bases and mechanisms underlying schizophrenia.

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